

ADVANCED SQL

DATE FUNCTIONS

Table 1 : Employees

emp-id	Name	hire date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1 : Add 6 months using DATEADD

```
SELECT emp-id, name, hire-date,  
DATEADD( Month, 6, hire-date )  
AS hire-plus-6-months ;
```

emp-id	Name	hire-date	hire-plus-6
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2 : Students

student_id	Name	DOB
101	Maya	2008-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Q2 : USE DATEDIFF to find age in days from dob to today's

```
SELECT student_id, name,
       DATEDIFF(days, '2008-08-10', '2026-04-16')
       DATEDIFF(days, '2004-12-01', '2026-04-16'),
       DATEDIFF(days, '2006-03-15', '2026-04-16')
       AS age-in-days
FROM Students;
```

student_id	Name	age-in-days
101	Maya	7554
102	Ethan	7806
103	Sienna	7337

Table 3 : Events

event-id	event-name	event-date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Q3 : Find how many days left till each event using DATEDIFF

```

SELECT event-id, event-name,
       DATEDIFF(days, '2024-06-15', '2025-09-01')
       DATEDIFF(days, '2025-09-01', '2025-01-20')
       AS days_remaining
FROM Events ;

```

event-id	event-name	days_remaining
1	Seminar	0
2	Workshop	443
3	Hackathon	219

Table 4 : Invoices

invoice_id	issue_date	due_date
S01	2025-03-10	2025-03-25
S02	2025-04-01	2025-04-15
S03	2025-04-10	2025-04-20

Q4: Difference between issue_date and due_date

```
SELECT *
  DATEDIFF (days, '2025-03-10', '2025-03-25'),
  DATEDIFF (days, '2025-04-01', '2025-04-15'),
  DATEDIFF (days, '2025-04-10', '2025-04-20')
AS days_between
FROM Invoices ;
```

Invoice_id	issue_date	due_date	day_between
S01	2025-03-10	2025-03-25	15
S02	2025-04-01	2025-04-15	14
S03	2025-04-10	2025-04-20	10

Table S: Courses

course_id	Name	start-date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Q5 : Format start-date as 'Month YYYY'
using TO-CHAR

```
SELECT course_id, name,
       TO_CHAR(start-date, 'Month YYYY')
       AS formatted-date
FROM Courses;
```

Course_id	Name	formatted-date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6 : Memberships

member id	start-year	start-month	start-day
1	2023	5	10
2	2022	11	25

Q6 : Create full date using dates-from-part

```
SELECT member-id,
DATE_FROM_PART(2023, 5, 10),
DATE_FROM_PART(2022, 11, 25) AS
full-start-date
FROM Memberships;
```

member-id	full-start-date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

sub-id	plan	renewal-date
11	Basic	2025-01-01
12	Premium	2025-03-15

Q7: Add 1 year using DATEADD

```
SELECT sub-id, plan,  
       DATEADD (YEAR, 1, renewal-date)  
AS extended-renewal-date  
FROM Subscriptions;
```

sub-id	Plan	extended-date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8 : Orders

order-id	order-date
1001	2025-04-15
1002	2025-04-10

Q8: Show current date and difference from order

```
SELECT CURRENT-DATE AS today-date,  
       order-id, order-date  
       DATEDIFF (current-date, order-date)  
       AS days-since-order  
FROM Orders ;
```

order-id	order-date	days-since-order
1001	2025-04-15	366
1002	2025-04-10	371

Table 9 : Trainings

training-id	topic	training-date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Q9 : Extract the year from training-date

```
SELECT training-id , topic  
       DATE_PART (YEAR, training-date)  
       AS training_year  
FROM   Trainings ;
```

training-id	topic	training-year
1	Safety	2025
2	Compliance	2025

Table 10 : Blog Posts

Post-id	title	published-on
1	SQL tips	2025/04/01 10:15:00
2	Data cleaning	2025/04/12 16:45:00

Q10: Extract hour and minute

```

SELECT post-id, title
       DATE_PART (hour, published-on) AS
       hour_published,
       DATE_PART (minute, published-on) AS
       minute_published
FROM   Blog-Posts ;

```

Post-id	title	hour- published	minute- published
1	SQL tips	10	15
2	Data Cleaning	16	45

Table 11 : Drivers

driver_id	license_expiry
301	2025-08-10
302	2023-12-31

Q11: Calculate days left until the license expires using DATEDIFF

```
SELECT driver_id, expiry license_expiry,  
DATEDIFF (current_date, license_expiry)  
AS days_left  
FROM : DRIVERS ;
```

driver_id	license_expiry	days_left
301	2025-08-10	250
302	2023-12-31	810

Table 12 : Messages

message_id	sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Q12: Display the current timestamp and calculate seconds since the message was sent

```
SELECT message_id, sent_timestamp,
CURRENT_TIMESTAMP AS current_timestamp,
TIMESTAMPDIFF(SECOND, sent_timestamp,
CURRENT_TIMESTAMP) AS seconds_since_sent
FROM messages;
```

message_id	sent_timestamp	current_timestamp	seconds_since
1	2025-04-19 09:32	2026-04-29	32 416 033
2	2025-04-18 23:59	2026-04-29	32 450 401

Table 13 : Returns

returns-id	return-date
901	2025-04-05
902	2025-04-01

Q13: Add 15 days to return-date

```
SELECT return-id, return date,  
       DATEADD (days, 15, return-date )  
       AS restock-date  
FROM returns ;
```

returns-id	return-date	restock-date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14 : Assignments

Assign_id	Assigned_on
1	2025-03-01
2	2025-03-05

Q14 : Convert assigned_on to DATE
using TO_DATE

```
SELECT assign_id,  
       TO_DATE (Assigned_on 'YYYY-MM-DD')  
AS assigned-on-date  
FROM Assignments ;
```

Assign_id	assigned-on-date
1	2025-03-01
2	2025-03-05

Table 15 : Meetings

meeting-id	schedule-time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

```
SELECT meeting-id,  
       TO_CHAR(scheduled-time, 'Month DD'  
              'YYYY at HH:MI PM')  
AS formatted-meeting-time  
FROM meetings;
```

meeting-id	formatted-meeting-time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM